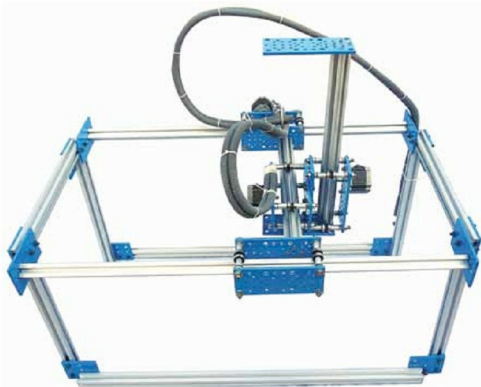


A Truly Modular Robotic Kit

Always wanted to get into the exciting field of robotics but didn't know where to start? Let us introduce you to one of the most innovative robotic kits of the decade—it could very well be described as the Lego set of the engineering community



ASHWIN GOPINATH

Remember your childhood, when a bucket of Lego bricks was like having infinite ammo for your imagination? Growing up doesn't mean you have to give up those moments. In fact, you can now use your imagination and make stuff that is much more complex and has far greater capabilities. In this article, we discuss an interesting kit that will allow you to create a myriad of robotic structures to perform various functions.

Merlyn TRN-1

At its simplest, it is a robotic kit that helps you design up to 28 different types of fundamental robots, be it a

linear moving robot, rotary robotic arm or even a robot with cylindrical axis. If you dig beneath the surface a little, it is a completely modular, electronic, rugged kit that not only allows you to create robots but also helps you add new features to them. This one kit allows a huge array of possibilities to create, control and reconfigure robots—whatever be the need.

It is basically a gadget, the parts of which can be attached together to create any kind of motion. You can add your regular devices to this gadget. The best part is that you can use it to perform functions required for industrial automation—such is the flexibility of this device.

This kit is the brainchild of Biju Ronnie Varkey, owner of Designs and

Projects Development (DPD), Patiala. A robotics enthusiast himself, Varkey worked on this product for two long years, making sure that it is built and functions exactly the way it is meant to be.

On asking what drove this idea in his mind, Varkey says, "While teaching robotics at engineering colleges, I noticed that there was nothing available at hand to teach students in a way that they would not only remember but also enjoy learning robotics. The most I saw anywhere was a couple of microcontrollers, line follower robots and the like. I sincerely felt that this was a huge waste of engineering talent and hence I started working on this project two years ago by designing and then fine-tuning the initial design according to the feedbacks I received from students and teachers, who are as much responsible for the final design as anyone who worked on it."

"Apart from generating an interest in robotics, I also wanted students, who come out having learnt the basics of robotics, to be able to say that they had an employable education," he adds.

Features of Merlyn TRN-1

The kit is basically divided into three parts: electronics, which includes microcontrollers, motors and drivers for motors; mechanical part, which comprises all the axes; and software segment, which controls microcontrollers used in the kit.

The kit uses a stepper motor for movement along all the axes, including the rotary axis that has a geared set-up. The motor used is a 2A stepper motor with independent drivers for each of the motors. These drivers are con-